

Practical Assignment 1 (AI DS), B-tech 5th Semester, III Year, Solve Using the Python

Q.1

- a) Assume a simple pair of fair dice being thrown. Generate the probability distribution for all the possible states of the sum of the two dice being thrown.
- b) Assume that a person stands at position 0 at time 0. He tosses a coin. He will move one step forward if the coin lands heads, one step backwards if it lands tails. Generate the probability distribution for all the states after 10-coin tosses.
- c) After 10-coin tosses, what is the probability that the person is at positions
 - i) 0
 - ii) 5
 - iii) 10

Q.2 A drone starts at position 0. Every second, it experiences wind disturbance.

It moves one step forward if the wind pushes forward.

It moves one step backward if the wind pushes backward.

Assume the wind direction is equally likely in each second.

- a) Generate the probability distribution for all possible positions of the drone after 8 seconds.
- b) Plot the probability distribution.
- c) Find the probability that the drone is at
 - i) Position 0
 - ii) Position 4
 - iii) Position 8